UNIVERSITY OF THE ARMED FORCES - ESPE

Computer Science Department

Object Oriented Programming

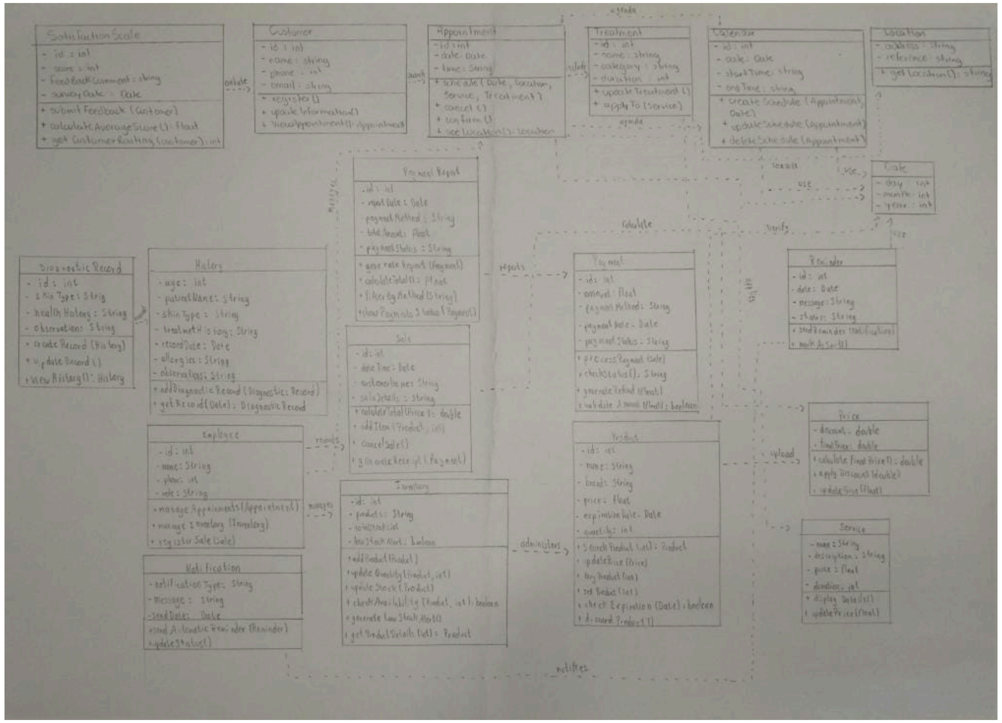
36863

Homework06 - ProyectClassModeling

LASCANO, JORGE EDISON

October 14, 2026

MICKEY
EST
1R26
irates
Y BETTER



Satisfaction Scale
- id : int
- score : int
- FeedBackComment : string
- surveyDate : Date
+ submitFeedback (Customer)
+ calculateAverageScore () : Float
+ getCustomerRating (Customer) : int
update
Customer
- id : int
- name : string
- phone : int
- email : string
+ register ()
+ update Information ()
+ viewAppointment () : Appointment
search
Appointment
- id : int
- date : Date
- time : String
+ schedule (Date, Location, Service, Treatment)
+ cancel ()
+ confirm ()
+ seeLocation () : Location
include
agenda
Treatment
- id : int
- name : string
- category : string
- duration : int
+ updateTreatment ()
+ applyTo (Service)
agenda
Calendar
- id : int
- date : Date
- startTime : string
- endTime : string
+ createSchedule (Appointment, Date)
+ updateSchedule (Appointment)
+ deleteSchedule (Appointment)
Location
- address : String
- reference : string
+ getLocation () : string
use
use
Date
- day : int
- month : int
- year : int
manages
Payment Report
- id : int
- reportDate : Date
- paymentMethod : String
- totalAmount : float
- paymentStatus : String
+ generateReport (Payment)
+ calculateTotal () : float
+ filterByMethod (String)
+ showPaymentsStatus (Payment)
reports
calculate
verify
use
Diagnostic Record
- id : int
- skinType : String
- healthHistory : String
- observation : String
+ createRecord (History)
+ updateRecord ()
+ viewHistory () : History
History
- age : int
- patientName : string
- skinType : String
- treatmentHistory : String
- recordDate : Date
- allergies : String
- observations : String
+ addDiagnosticRecord (Diagnostic : Record)
+ getRecord (Date) : Diagnostic Record
Sale
- id : int
- dateTime : Date
- customerName : String
- saleDetails : String
+ calculateTotal (Price) : double
+ addItem (Product, int)
+ cancelSale ()
+ generateReceipt (Payment)
Payment
- id : int
- amount : float
- paymentMethod : String
- paymentDate : Date
- paymentStatus : String
+ processPayment (Sale)
+ checkStatus () : String
+ generateRefund (float)
+ validateAmount (float) : boolean
Reminder
- id : int
- date : Date
- message : String
- status : String
+ sendReminder (Notification)
+ markAsSent ()
Employee
- id : int
- name : String
- phone : int
- role : String
+ manageAppointments (Appointment)
+ manageInventory (Inventory)
+ registerSale (Sale)
records
manages
Inventory
- id : int
- products : String
- totalStock : int
- lowStockAlert : boolean
+ addProduct (Product)
+ updateQuantity (Product, int)
+ updateStock (Product)
+ checkAvailability (Product, int) : boolean
+ generateLowStockAlert ()
+ getProductDetails (int) : Product
administers
Product
- id : int
- name : String
- brand : String
- price : float
- expirationDate : Date
- quantity : int
+ searchProduct (int) : Product
+ updatePrice (Price)
+ buyProduct (int)
+ sellProduct (int)
+ checkExpiration (Date) : boolean
+ discardProduct ()
upload
Price
- discount : double
- finalPrice : double
+ calculateFinalPrice () : double
+ applyDiscount (double)
+ updatePrice (float)
Service
- name : String
- description : String
- price : float
- duration : int
+ displayDetails ()
+ updatePrice (float)
Notification
- notificationType : String
- message : String
- sendDate : Date
+ sendAutomaticReminder (Reminder)
+ updateStatus ()
notifies